SOC ANALYST MASTER ROADMAP

A comprehensive 19-phase curriculum covering Security Operations Center defenses, SIEM analysis, threat hunting, and incident response.

PHASE 1 Computer Fundamentals

1. Computer Hardware Fundamentals
2. Operating System Fundamentals
3. Windows Architecture
4. Linux Architecture
5. Windows File Systems
6. Linux File Systems
7. Processes and Threads
8. Services and Daemons
9. User Accounts and Permissions
10. Windows Command Line
11. Linux Terminal Commands
12. System Administration Basics

PHASE 2 Networking Fundamentals

1. Networking Basics
2. OSI Model
3. TCP/IP Model
4. Network Topologies
5. IPv4 and IPv6
6. Subnetting
7. MAC Addressing
8. ARP Protocol
9. TCP and UDP
10. ICMP Protocol
11. DNS & DHCP
12. NAT & Routing/Switching
13. VLANs & Common Ports
14. Network Protocols
15. Firewalls, VPN & Proxies

PHASE 3 Windows Security

1. Windows Security Architecture
2. Windows Registry
3. Windows Event Logs & Event Viewer
4. Windows Defender & Firewall
5. Windows Authentication
6. Active Directory Fundamentals
7. Group Policy & Account Management
8. NTLM & Kerberos Authentication
9. PowerShell Fundamentals
10. Windows Security Auditing & Sysmon

PHASE 4 Linux Security

1. Linux Security Architecture
2. Linux User Management & Permissions
3. Linux Process & Service Management
4. Linux Authentication & SSH Security
5. Linux Firewall & System Logs
6. Syslog & Journalctl
7. Linux Auditing & Hardening

PHASE 5 Cybersecurity Fundamentals

1. Information Security Fundamentals
2. CIA Triad & AAA Security Model
3. Threats, Vulnerabilities & Controls
4. Risk Management & Attack Vectors
5. Malware, Ransomware & Phishing
6. Social Engineering & Insider Threats
7. Advanced Persistent Threats (APT)
8. Indicators of Compromise (IOC) & Attack (IOA)

PHASE 6 Security Monitoring

1. Security Monitoring Fundamentals
2. Network & Endpoint Monitoring
3. Server, Log & Event Monitoring
4. User Activity & Auth Monitoring
5. Network Traffic Monitoring
6. Real-Time Threat & Alert Monitoring
7. Security Event Correlation

PHASE 7 Log Analysis

1. Log Fundamentals
2. Windows Event & Linux System Logs
3. Firewall, DNS & DHCP Logs
4. VPN, Proxy & Web Server Logs
5. Authentication & Endpoint Logs
6. Sysmon Logs Analysis
7. Log Correlation & Filtering
8. Log Investigation Procedures

PHASE 8 SIEM Fundamentals

1. SIEM Architecture & Components
2. Log Collection, Aggregation & Parsing
3. Log Normalization & Enrichment
4. Event Correlation & Alert Generation
5. Dashboards & SIEM Queries
6. Detection Rules & Use Cases
7. Threat Detection & Deployment

PHASE 9 SIEM Tools Stack

1. Splunk | **2. IBM QRadar** | **3. Microsoft Sentinel** | **4. Elastic Security** | **5. Wazuh** | **6. ArcSight** | **7. LogRhythm** | **8. Graylog**

PHASE 10 Threat Intelligence

1. Threat Intelligence Fundamentals
2. Strategic, Tactical & Operational CTI
3. Technical Threat Intelligence
4. Threat Intelligence Lifecycle
5. IOC, IP, Domain & Hash Analysis
6. Threat Feeds Integration
7. MITRE ATT&CK Framework
8. Cyber Kill Chain & Diamond Model
9. Threat Actor Profiling

PHASE 11 Incident Detection & Response

1. Incident Response Fundamentals
2. Identification, Classification & Triage
3. Incident Containment & Eradication
4. System Recovery & Post-Incident Review
5. Incident Documentation & Reporting
6. Incident Response Lifecycle
7. Security Playbooks & Escalation

PHASE 12 Malware Analysis Fundamentals

1. Malware Fundamentals & Classification
2. Viruses, Worms & Trojans
3. Ransomware, Spyware & Rootkits
4. Keyloggers & File Hash Analysis
5. Static Malware Analysis
6. Dynamic Malware Analysis
7. Sandbox & Network Analysis
8. Malware Indicators Extraction

PHASE 13 Endpoint Detection & Response

1. Endpoint Security Fundamentals
2. EDR & XDR Architecture
3. Endpoint Telemetry & Process Monitoring
4. File Activity & Registry Monitoring
5. Endpoint Threat Detection
6. Behavioral Analysis & Threat Hunting
7. Endpoint Isolation & Alert Investigation

PHASE 14 Network Security Monitoring

1. NSM Fundamentals
2. Packet Analysis & Wireshark Sniffing
3. TCP Stream & DNS Traffic Analysis
4. HTTP & TLS Traffic Analysis
5. Network Intrusion Detection (IDS/IPS)
6. Suricata, Snort & Zeek Operation
7. Network Anomaly Detection

PHASE 15 Vulnerability & Risk Management

1. Vulnerability Management Basics
2. Assessment & Scanning (CVE, CVSS, CWE)
3. Patch Management & Misconfigurations
4. Asset Management & Risk Assessment
5. Risk Prioritization & Remediation
6. Compliance Monitoring

PHASE 16 Threat Hunting

1. Threat Hunting Fundamentals
2. Hypothesis-Based & IOC-Based Hunting
3. Behavioral & Network Threat Hunting
4. Endpoint & Log-Based Threat Hunting
5. MITRE ATT&CK-Based Hunting
6. Hunting Queries & Anomaly Detection
7. APT Detection & Hunting Reports

PHASE 17 Digital Forensics Fundamentals

1. Digital Forensics Fundamentals
2. Evidence Collection & Preservation
3. Chain of Custody Maintenance
4. Disk & Memory Forensics
5. Windows & Linux Forensics
6. Network & Browser Forensics
7. Timeline & File System Analysis
8. Forensic Reporting

PHASE 18 SOC Operations & Professional Skills

1. SOC Architecture & Team Structure
2. SOC Analyst Roles (L1, L2, L3)
3. Alert Triage & True/False Positive Analysis
4. Security Ticket Management & SLAs
5. Incident Escalation & Shift Handover
6. SOC Documentation & Security Reporting
7. SOC Metrics and KPIs

PHASE 19 Practical SOC Labs & Hands-on Investigations

1. Windows Event Log Analysis Lab
2. Linux Log Analysis Lab
3. Sysmon Configuration Lab
4. Splunk SIEM Lab
5. Microsoft Sentinel Lab
6. Wazuh SIEM Lab
7. Brute-Force Detection Lab
8. Failed Login Investigation Lab
9. Malware Alert Investigation Lab
10. Phishing Email Investigation Lab
11. Suspicious PowerShell Investigation Lab
12. Network Traffic Analysis Lab
13. IDS Alert Investigation Lab
14. Ransomware Incident Simulation
15. Incident Response Simulation
16. Threat Hunting Lab
17. Digital Forensics Lab
18. Complete SOC Investigation Project